CAUÃ FELYPE CAETANO REIS

Full Stack Software Engineer · Python (FastAPI), TypeScript & AI-Augmented Systems

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🔗 linkedin.com/in/cauafelypereis 💻 github.com/cauacreis 🌐 Portfolio (Vercel)

PROFESSIONAL SUMMARY

Results-oriented Full Stack Software Engineer with a proven track record of architecting, deploying, and maintaining high-throughput asynchronous backend systems with **Python (FastAPI, AsyncIO)**, scalable relational databases (**PostgreSQL**), and modern reactive frontends (**TypeScript, Vue.js, React/Next.js**). Lead architect and 24/7 operator of **SKZ Trading**, a live algorithmic trading SaaS platform orchestrating 11 distributed Python microservices (< 100ms latency, 99.9% uptime, automated webhooks, and zero-loss financial reconciliation). Co-author of a scientific research paper submitted to **IEEE Transactions on Software Engineering (IEEE TSE)** on Multi-Agent Systems (MAS/MoE) and Explainable AI. Advanced English speaker, certified in Agile (**Scrum SFC™**), and an enthusiastic pioneer of **AI-assisted engineering workflows** utilizing **Google Antigravity, Claude Code, Gemini CLI, and Coding Agents** to deliver resilient software with forensic attention to detail.

CORE TECHNICAL SKILLS

Backend & APIs
Python 3.11+, FastAPI, AsyncIO, aiohttp, RESTful APIs, WebSockets (real-time bidirectionality), Pydantic v2, Background Tasks & Worker Services, Microservices Architecture.

Frontend & Cloud Deployments
TypeScript, JavaScript, Vue.js, React, Next.js (App Router), Tailwind CSS, Vite, Vercel deployments, AWS, Docker, responsive UI/UX engineering.

Relational Databases & Schema Design
PostgreSQL (schema modeling, relational indexing, query performance tuning, ACID transactions), MySQL, Redis (caching & pub/sub), SQLite (Outbox pattern).

AI-Enthusiastic Workflow & Quality
Google Antigravity, Claude Code, Gemini CLI, Cursor, OpenAI Codex, Coding Agents, Playwright & Jest automated testing, Clean Code, SOLID, Scrum (SFC™ Certified).

PROFESSIONAL EXPERIENCE & PRODUCTION SYSTEMS

Full Stack Software Engineer & Lead Architect — SKZ Trading `24/7 Production` Jul 2026 – Present

High-throughput digital asset algorithmic trading SaaS integrated with real-time financial and partner APIs

- **High-Concurrency Python Microservices:** Architected, deployed, and maintain 11 asynchronous microservices built with Python (`asyncio`, `aiohttp`) managed via PM2 on Linux VPS, running continuous 20-second scans with sub-100ms API response latency.
- **FastAPI & RESTful API Delivery:** Engineered resilient API endpoints handling authentication, automated webhooks, data serialization, and real-time transaction processing.
- **PostgreSQL & Database Optimization:** Designed relational schema models, optimized SQL queries and indexing strategies, ensuring zero-loss transactional consistency across hundreds of automated trades.
- **Financial Pipeline & Anti-Loss Engine:** Implemented proprietary pricing algorithms with liquidity scoring and hard profit safeguards, automating split payouts via Mercado Pago webhooks with 99.9% uptime.
- **Security Hardening:** Enforced local server-side asymmetric/symmetric encryption (Ed25519/AES) adhering to OWASP Top 10 standards, eliminating exposed secrets and securing API payloads in transit.

KEY ENGINEERING PROJECTS & APPLIED RESEARCH

OrenAI — Enterprise AI Platform & Research Paper (IEEE TSE)

Distributed Systems / MAS

2025 – Present

- **Multi-Agent Architecture (MAS / MoE):** Architected a containerized distributed system running commodity-specific agent experts in Docker to accelerate mineral prospectivity mapping and derisk exploration.
- **Scientific Publication:** Co-authored a research paper submitted to **IEEE Transactions on Software Engineering**, leading formal architectural specifications, technical documentation, and system topology diagrams.
- **AI-Assisted Workflow:** Spearheaded the development lifecycle utilizing agentic coding tools (Google Antigravity, Claude Code, Gemini) to accelerate benchmarking, integration tests, and data pipeline validations.

GymFlow — Mobile-First SaaS Application (Deployed on Vercel)

Next.js / TypeScript / Vercel

Sep 2026

- **Modern Full Stack Deployment:** Built and shipped a high-performance web application utilizing Next.js/React, TypeScript, and Tailwind CSS, deployed on **Vercel** with optimized edge rendering.
- **Dynamic Security Passes:** Engineered a dynamic QR-code rotational token system with 30-second in-memory cycles, preventing fraud and unauthorized credential sharing.
- **AI & Payment Integration:** Connected Google Gemini API for intelligent fitness coaching and integrated Mercado Pago checkout for automated recurring billing.

Bodega do Vidigal — Offline-First POS & Kitchen Display System (KDS)

React Native / SQLite P2P

2026

- **Local-First P2P Networking:** Built an offline-first POS and KDS ecosystem communicating peer-to-peer over local mobile Wi-Fi (device hotspot) via WebSockets with < 10ms latency and zero cellular data usage.
- **Transactional Outbox Pattern:** Structured local SQLite storage with an atomic Outbox pattern, guaranteeing zero duplicate orders and complete state recovery upon unexpected restarts.
- **Automated Test Suite:** Created 37 comprehensive automated unit and end-to-end tests (Jest) validating order state machines, FIFO kitchen prioritization, and nightly auto-resets.

EDUCATION & CERTIFICATIONS

B.S. in Software Engineering — UniEVANGÉLICA

4th Semester | Expected: 2027/2028

- Evening classes — **100% available for full-time business hours (Monday – Friday)**.
- Vice-President of the CodeUp Academic Directory; Director of Arts & Tech Culture.

Desktop Developer Certification (C#, Java, Relational SQL, OOP, UML) — Instituto Mix

Oct 2022 – Jun 2024

- Practical coursework in Object-Oriented Programming, Relational Modeling, and Distributed System Fundamentals.
- **Certifications:** Scrum Fundamentals Certified (SFC™) · Technical Storytelling & System Documentation.
- **Languages:** Portuguese (Native) · English (Advanced — Fluent technical documentation, scientific writing, and verbal collaboration).